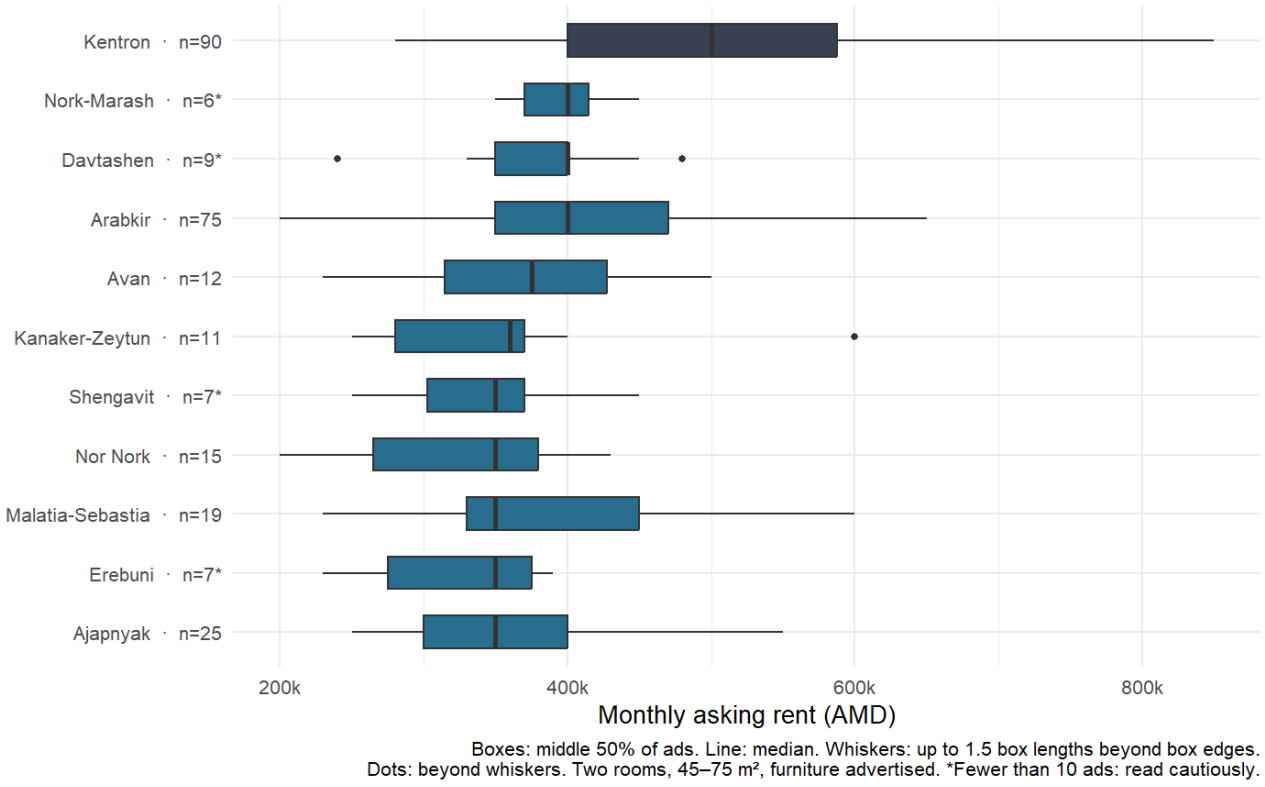
Does cheaper housing make life more expensive?

Yerevan · 5 November 2026

Lower rent survives the baseline comparison. Six charts compare real asking rents with modeled travel and winter bills.

Select a chart to open its full-size image.

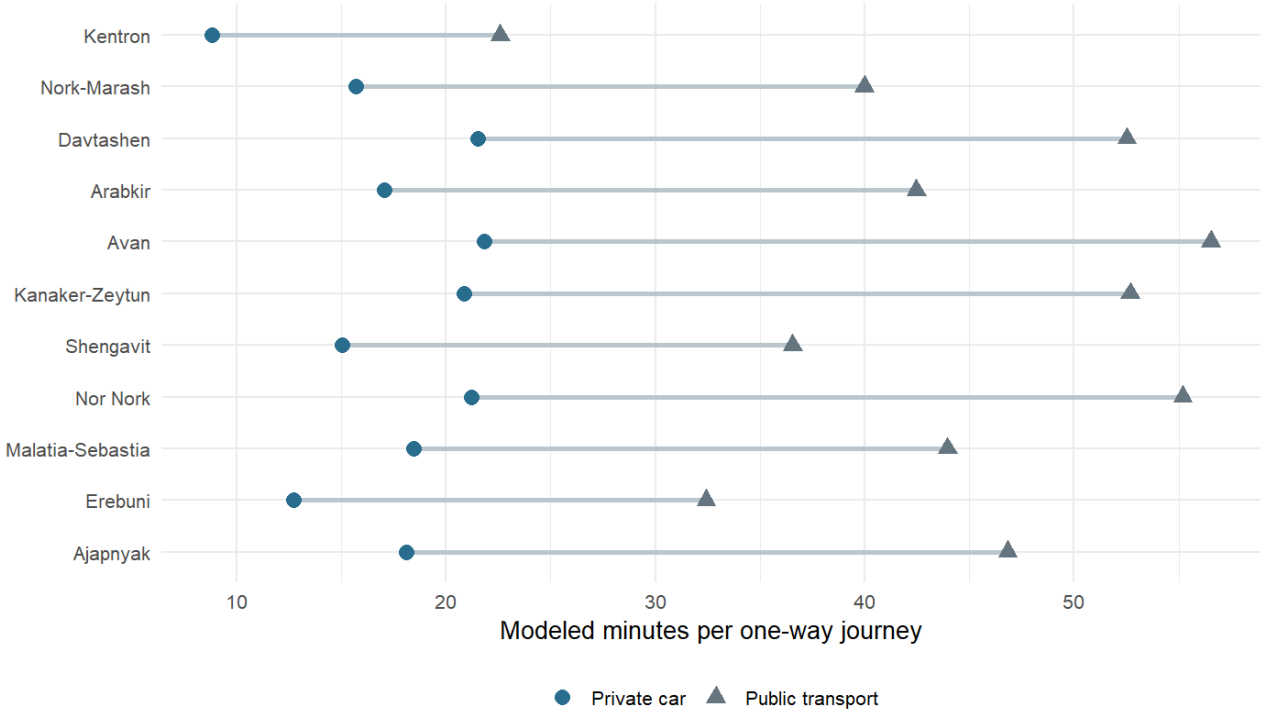
Where is rent cheaper?



Boxes: middle 50% of ads. Line: median. Whiskers: up to 1.5 box lengths beyond box edges.
Dots: beyond whiskers. Two rooms, 45–75 m², furniture advertised. *Fewer than 10 ads: read cautiously.

Kentron has the highest median asking rent. Its median is 500,000 AMD, compared with 350,000–400,000 AMD elsewhere in this sample. The boxes show overlapping distributions, so district alone does not determine a listing's price.

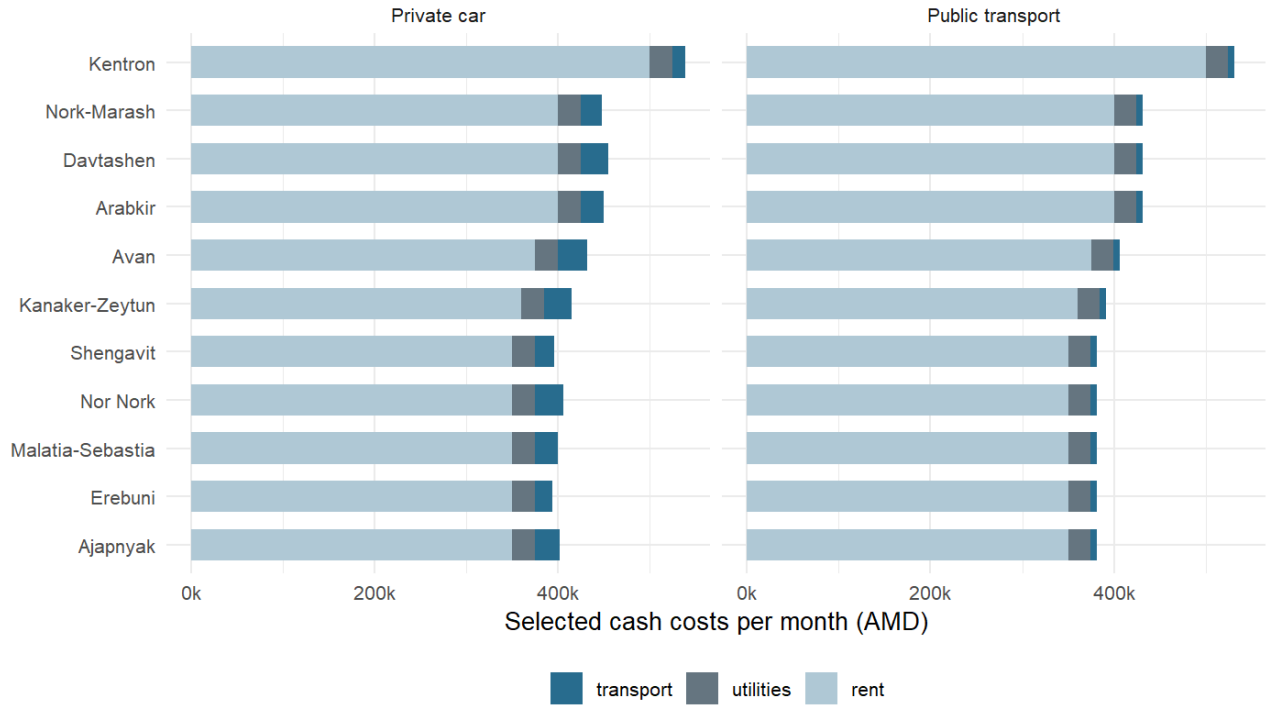
How much longer is the commute?



Each connecting line shows the difference between the two modeled travel modes.
Destination: Republic Square. Transit: 12 km/h + 15 min per leg; car: 1.5 × free-flow + 5 min per leg.

Public transport takes longer in these modeled comparisons. Longer connecting lines show a larger gap between the two modes. Kentron is closer to the shared destination. These are road-based scenarios—not timed journeys, live traffic, or actual bus routes.

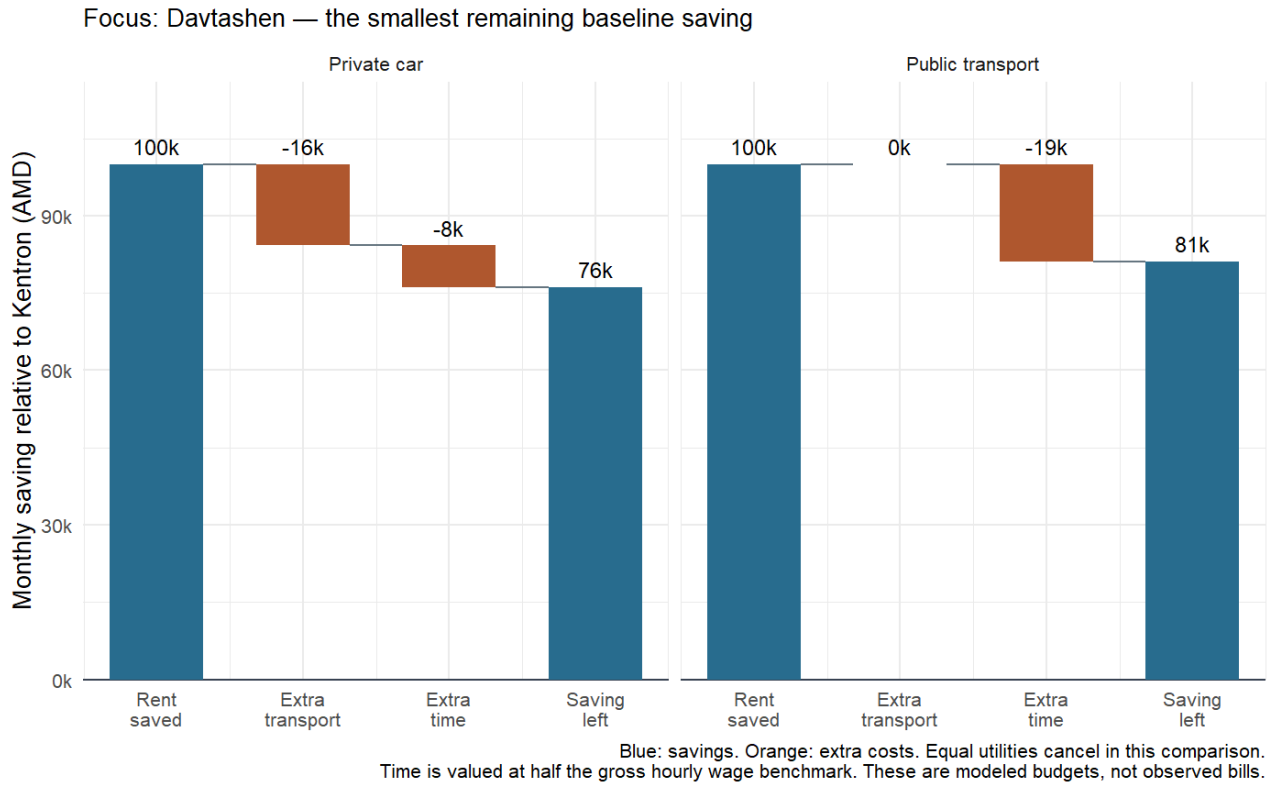
What does the monthly budget include?



Rent + modeled electricity, gas, water and commuting. One commuter, 22 return-commute days. Same winter consumption in every district; heating is included once through gas.

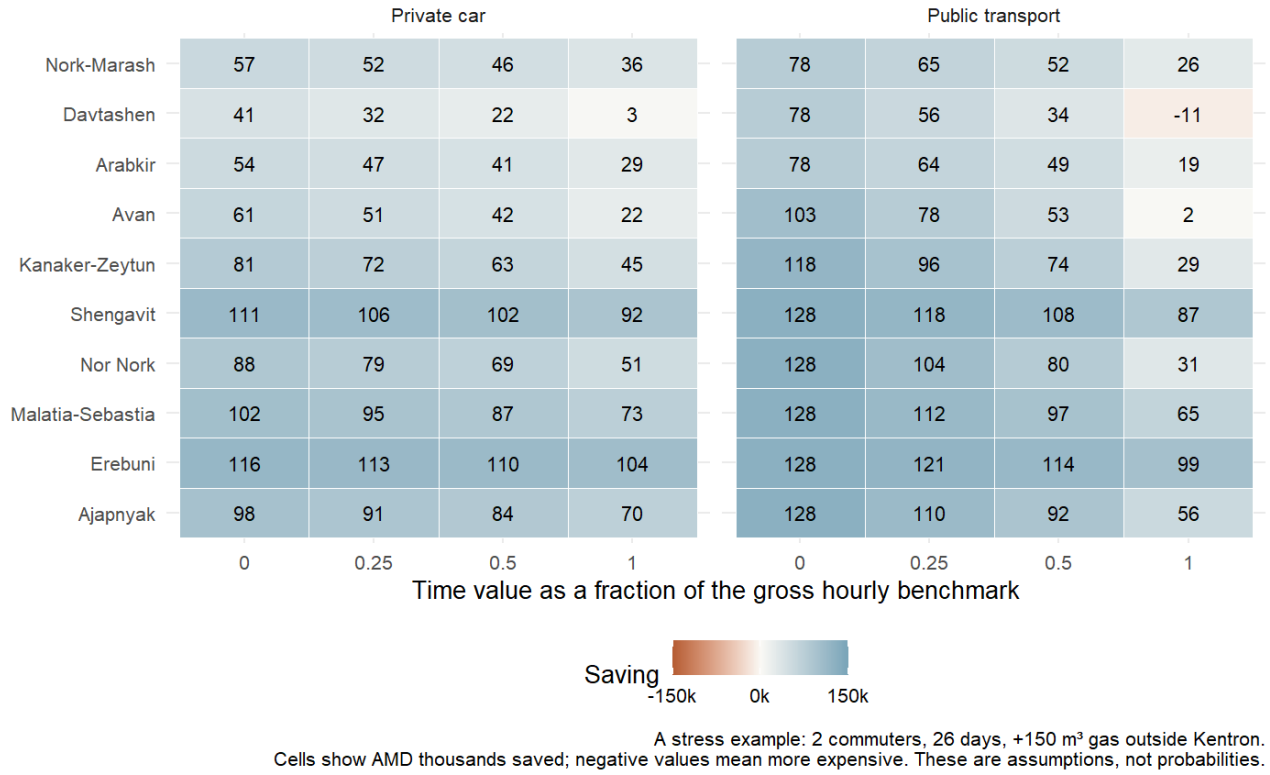
Rent is the largest part of the selected budget. Modeled winter utilities add 24,769 AMD everywhere. Public fares can be the same despite a longer journey; driving adds distance-dependent costs. Time is not a cash cost and is added in the next chart.

How much saving remains after valuing time?



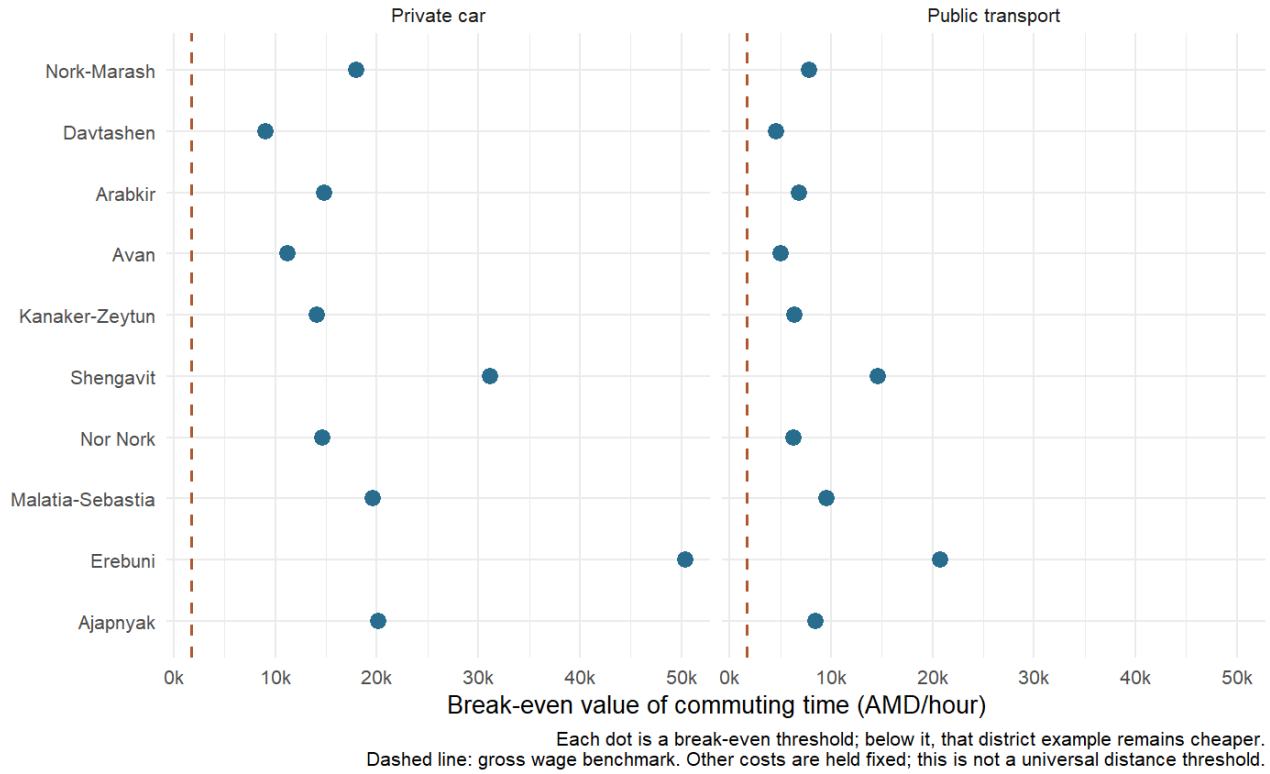
Even the most exposed baseline example retains a saving. The waterfall follows Davtashen, selected because it contains the smallest remaining saving. All 20 district–mode examples remain cheaper than Kentron at about 861 AMD/hour for time. These results apply to the stated sample and assumptions.

When can the answer change?



More commuting and heating can erase the saving in some cases. The chosen grid contains 2 more-expensive cases. The chart shows one heavy-commuting winter example; it does not estimate how often households face those conditions. Two-car cases assume separate journeys and vehicles.

When would the saving disappear?



The answer depends on how you value extra time. The lowest positive threshold is about 4,555 AMD/hour, above the 1,722 AMD/hour gross benchmark used here. This is a conditional calculation, and some district samples are small.

What should a renter take from this?

In the baseline model, the rent saving is large enough to cover the selected extra costs. A renter should still compare a real quote with their own commute, heating needs and value of time. The stress chart shows why the conclusion should remain conditional.

The evidence does **not** support a blanket claim that cheaper noncentral housing always becomes more expensive. It also does not prove that every noncentral apartment is cheaper overall.

Assumptions and sources

Read the household and travel assumptions

The household example

Two adults, one commuter, 22 return-commute days per 30-day month. Winter consumption is assumed to be 150 daytime plus 50 nighttime kWh, 100 m³ gas and 8 m³ metered water. Equal consumption is used across districts. These are not observed bills.

Public transport uses the cheaper of single rides and one 30-day pass per commuter. Car costs assume 60 AMD/km plus 10,000 AMD monthly parking per commuter, excluding full vehicle ownership. Public time uses car-road distance at 12 km/h plus 15 minutes per leg; car time uses 1.5 times free-flow duration plus 5 minutes per leg. Neither is an observed peak journey or bus itinerary.

Time is valued at half of Armstat's 2025 national gross monthly wage divided by 176 assumed paid hours. Zero time value gives the direct cash-cost result. Longer-pass commitments and social discounts are outside the baseline.

How the district budgets work

Each example combines its sample-median asking rent with median round-trip road inputs. Those medians may come from different apartments. Adding these components creates a modeled district budget—not the median of actual household bills. Kentron has its own nonzero commute to Republic Square.

Rent, modeled electricity/gas/water and commuting are included. Food, internet, building fees, moving costs, deposits and full car ownership are excluded. Heating is counted once through gas.

Read about the sample, statistics and checks

Data and limitations

The four CSVs in `data/` are the complete inputs. `rentals.csv` is already cleaned: 276 native-AMD, two-room, 45–75 m² advertisements with furniture advertised and usable route inputs. The original 949-record extraction, exclusion evidence and route replies are preserved in `_archive/previous-version/`.

This is one agency's catalogue, retrieved on 5 November 2026—not random sampling or completed leases. Availability and last price-update dates are unverified. Condition and building information are incomplete, and district labels/coordinates may be approximate. These comparisons are not quality-adjusted or causal. Outside Kentron is not outside the Yerevan municipality; Abovyan and Masis are not compared.

Boxes show listing quartiles, and whiskers extend to values within 1.5 box lengths; dots show values beyond them. These are not confidence intervals. We do not manufacture population p-values from convenience advertisements or scenario rows. The graphs communicate descriptive statistics and sensitivity to assumptions.

The original broader checks included faster/slower travel, transfers, a narrower size band and alternative aggregation. Their evidence remains archived; the simplified report retains the same primary sample, baseline calculations and full household/heating/time grid.

Reproduce the report

Open `YerevanHousing.Rproj`, open this Rmd, then click **Knit**. Both HTML and Unicode-capable PDF go into `outputs/`; the six chart images go into `figures/`. All calculations run here using local CSVs. No development scripts, downloads or new package installations are required during knitting. HTML code is folded; use its Code buttons to reveal the chart code. The matching Word file explains every chunk.

AI assistance was used for research, coding, writing and verification. The student should understand the work and confirm the instructor's submission and AI policies. Existing package-build warnings from newer R patch versions are a known environment limitation; the report is checked by actual execution.

Open the reference list

References

- WSE.AM. WSE.AM rental catalogue and public property API. Snapshot 2026-11-05. Accessed 2026-11-05.
- WSE.AM. Public property API access rules. Accessed 2026-11-05. Accessed 2026-11-05.
- Yerevan Municipality. Unified public transport ticket comparison. 2025 onward. Accessed 2026-11-05.
- Electric Networks of Armenia. Household electricity tariffs. 2022 rates with 2025 amendments. Accessed 2026-11-05.

- Public Services Regulatory Commission. Natural gas tariff decision. 2022-03-01. Accessed 2026-11-05.
- Public Services Regulatory Commission. Resident gas tariffs unchanged from August 2026. 2026. Accessed 2026-11-05.
- Veolia Djur. 2026 drinking water tariff. 2025-11-26. Accessed 2026-11-05.
- Veolia Djur. Water tariff units and metering FAQ. 2025 tariff description. Accessed 2026-11-05.
- Armstat. SDG8.5.1.a average monthly nominal wages and metadata. Data updated 2026-06-10; observation 2025. Accessed 2026-11-05.
- Project OSRM. OSRM HTTP API table service. v26.5.0 documentation. Accessed 2026-11-05.
- OpenStreetMap contributors. OpenStreetMap data attribution and license. Accessed 2026-11-05. Accessed 2026-11-05.
- Habet Madoyan, American University of Armenia. DS116 lecture materials. Repository commit 1a74f34584c85b08b0f9a87d055b663d209df9de. Accessed 2026-11-05.
- ggplot2 authors. ggplot2 reference. Documentation accessed 2026-11-05. Accessed 2026-11-05.
- dplyr authors. dplyr reference. Documentation accessed 2026-11-05. Accessed 2026-11-05.
- Yihui Xie and contributors. knitr documentation. Documentation accessed 2026-11-05. Accessed 2026-11-05.

Network estimates use OpenStreetMap contributors' data through OSRM. Source access does not establish an open redistribution license for the agency catalogue.